Module 2: Portfolio Milestone

Start Assignment

- Points 10
- Submitting a file upload

Portfolio Milestone (10 points)

Important! Read First

Choose one of the following two assignments to complete this week. Do not do both assignments. Identify your assignment choice in the title of your submission. Review the **Module 8 Portfolio Project (https://csuglobal.instructure.com/courses/123286/assignments/2269095)** instructions.

Option #1: Using C Programming language

In your Linux installation, use Bash, or a Linux Shell of your choice, to do the following:

1. Generate a random number: *echo $RANDOM*
2. Output a random number into a file called file1.txt: *echo $RANDOM > file1.txt*
3. *Append another random number to the end of this file: echo $RANDOM >> file1.txt*
4. Remove file1.txt: *rm file1.txt*
5. Create a script called numbers.sh, that does this one thousand (1000) times, using the “for” loop.
6. Make the script executable: *chmod +x numbers.sh*
7. Run the script: *./numbers.sh*

If you're not familiar with Linux Shell scripting, review the following videos:

Simpson, S. (2025, July 1). ***Learning bash scripting*** *(https://www.linkedin.com/learning-login/share?account=2245842&forceAccount=false&redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Flearni bash-scripting-26210777%2Flearning-bash-scripting%3Ftrk%3Dshare_video_url%26shareId%3DR1bJqP8lRpSGJhY29zt8Xg%253D%253D)* [Video]. LinkedIn Learning.

Simpson, S. (2025, July 1). ***Working with while and until loops*** *(https://www.linkedin.com/learning-login/share?account=2245842&forceAccount=false&redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Flearni*

bash-scripting-26210777%2Fworking-with-while-and-until-loops%3Ftrk%3Dshare_video_url%26shareId%3DQFyE0luaToeV9U33HNlp1g%253D%253D) [Video]. LinkedIn Learning.

You should now have a file called file1.txt containing 1,000 lines, with each line being a random number.

Create a C program to perform this task, to create file2.txt. You should now have 2 files: file1.txt & file2.txt, each containing 1,000 lines, with each line being a random number.

Do a word and line count on these programs, scripts, and text files (feel free to create a new folder(s) to store these, if you prefer a certain level of organization.)

*wc **

Take a screenshot of the files and their word/line counts and submit to the instructor.

Option #2: Using Python Programming language

In your Linux installation, use Bash, or a Linux Shell of your choice, to do the following:

1. Generate a random number: *echo $RANDOM*
2. Output a random number into a file called file1.txt: *echo $RANDOM > file1.txt*
3. *Append another random number to the end of this file: echo $RANDOM >> file1.txt*
4. Remove file1.txt: *rm file1.txt*
5. Create a script called numbers.sh, that does this one thousand (1000) times, using the “for” loop.
6. Make the script executable: *chmod +x numbers.sh*
7. Run the script: *./numbers.sh*

If you are not familiar with Linux Shell scripting, review the following videos:

Simpson, S. (2025, July 1). ***Learning bash scripting*** *(https://www.linkedin.com/learning-login/share?account=2245842&forceAccount=false&redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Flearni bash-scripting-26210777%2Flearning-bash-scripting%3Ftrk%3Dshare_video_url%26shareId%3DR1bJqP8lRpSGJhY29zt8Xg%253D%253D)* [Video]. LinkedIn Learning.

Simpson, S. (2025, July 1). ***Working with while and until loops*** *(https://www.linkedin.com/learning-login/share?account=2245842&forceAccount=false&redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Flearni bash-scripting-26210777%2Fworking-with-while-and-until-loops%3Ftrk%3Dshare_video_url%26shareId%3DQFyE0luaToeV9U33HNlp1g%253D%253D)* [Video]. LinkedIn Learning.

You should now have a file called file1.txt containing 1,000 lines, with each line being a random number.

Create a Python program to perform this task, to create file2.txt. You should now have 2 files: file1.txt & file2.txt, each containing 1,000 lines, with each line being a random number.

Do a word and line count on these programs, scripts, and text files (feel free to create a new folder(s) to store these, if you prefer a certain level of organization.)

*wc **

Take a screenshot of the files and their word/line counts and submit to the instructor.

CSC507_M02_PM

Criteria	Ratings				Pts
Completion of Bash Script Tasks	**3 pts** **Meets Expectation** All required Bash commands and steps (generate, append, remove, loop, make executable, and run) are correctly executed and documented.	**2 pts** **Approaches Expectation** Most steps completed correctly with minor omissions or syntax errors.	**1 pts** **Below Expectation** Some tasks attempted but with major errors or missing multiple required steps.	**0 pts** **Limited Evidence** Little or no evidence of correct Bash scripting or task completion.	3 pts
Implementation of C or Python Program	**3 pts** **Meets Expectation** Correctly creates file2.txt with 1,000 random numbers; program runs without error and mirrors Bash functionality.	**2 pts** **Approaches Expectation** Program runs with minor errors or produces incomplete output (e.g., missing lines or formatting issues).	**1 pts** **Below Expectation** Program attempts the task but fails to produce correct or consistent output.	**0 pts** **Limited Evidence** No evidence of a working C or Python program.	3 pts
Output Verification & Organization	**2 pts** **Meets Expectation** Demonstrates accurate use of wc *; includes screenshot showing correct word/line counts and organized file structure.	**1.5 pts** **Approaches Expectation** Includes screenshot or output with minor inaccuracies or missing files.	**1 pts** **Below Expectation** Output is incomplete, disorganized, or partially missing.	**0.5 pts** **Limited Evidence** No screenshot or output evidence provided.	2 pts
Documentation & Clarity of Submission	**2 pts** **Meets Expectation** Submission clearly identifies assignment option (C or Python),	**1.5 pts** **Approaches Expectation** Submission identifies option but has minor	**1 pts** **Below Expectation** Submission lacks clear identification	**0.5 pts** **Limited Evidence** Submission is unclear, mislabeled, or incomplete.	2 pts

Criteria	Ratings				Pts
	includes labeled files, and is well-organized.	clarity or labeling issues.	or has confusing structure.		
Total Points: 10					